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ORIGINAL ARTICLE

Preoperative Exposure to Sodium-Glucose Cotransporter-2 Inhibitors and Acute Kidney Injury After Cardiac Surgery: A Retrospective Single-Centre Cohort Study With Overlap Weighting Propensity Score Analysis

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Background & Aims This study aims to investigate the impact of preoperative sodium-glucose cotransporter 2 inhibitor (SGLT2i) exposure on cardiac surgery-associated acute kidney injury. This is a single-centre retrospective cohort study with a propensity score weighting analysis for causal inference. The study was set in a cardiovascular and thoracic centre of a teaching hospital (France), between January 2022 and February 2024.

Method Participants were adults who had undergone cardiac surgery with cardiopulmonary bypass and had ≥1 of the following indications for long-term treatment with SGLT2i: type 2 diabetes with high cardiovascular risk; heart failure with a left ventricular ejection fraction <50%; chronic kidney disease with a glomerular filtration rate <60 mL/min/1.73 m^2. Preoperative long-term SGLT2i medication versus no preoperative long-term SGLT2i medication, whether or not the treatment was discontinued during the preoperative period.

Results The primary end point was cardiac-associated acute kidney injury Stage ≥1 in the 7 days after surgery as defined by the creatinine variation (Kidney Disease: Improving Global Outcomes criteria). Causal inference analysis was performed with overlap weighting propensity score analysis. A total 509 patients were included (117 with preoperative exposure to SGLT2i, 65 patients without preoperative treatment discontinuation). Preoperative exposure to SGLT2i was not associated with a significant reduction in cardiac surgery-associated acute kidney injury (35% vs 30%, relative risk 1.12; 95% confidence interval 0.80–1.55).

Conclusions Preoperative exposure to SGLT2i was not associated with a reduction in cardiac surgery-associated acute kidney injury in the present study, which was underpowered to rule out a potential clinically significant effect.

Keywords Sodium-glucose cotransporter-2 inhibitors • Cardiac surgery-associated acute kidney injury • Cardiac surgery

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Introduction

Long-term treatment with sodium-glucose cotransporter-2 inhibitors (SGLT2is) improves outcomes in type 2 diabetes [1], chronic and acute heart failure [2], and in chronic kidney disease [3,4]. Although there was initial concern of the risk of acute kidney injury due to the physiological decline in glomerular filtration rate (GFR) observed with the introduction of SGLT2i, several studies have shown a reduction in the incidence of this in patients receiving long-term SGLT2i therapy [5–7]. SGLT2i have also been reported to be associated with a lower incidence of chronic kidney disease after acute kidney injury [8], and there is a scientific rational that SGLT2i may be nephroprotective through multiple pathways [9]. However, it is recommended to discontinue SGLT2i in the preoperative period to reduce the risk of euglycaemic ketoacidosis [10,11]; although potentially serious and difficult to diagnose, this complication is rare and, to our knowledge, no fatal case has been reported in the literature [12]. Cardiac surgery-associated acute kidney injury is common and is associated with worse short- and long-term outcomes. The causes of renal insults during, and in the days following, cardiac surgery are multiple and intricate, leading to direct nephrotoxicity and hypoperfusion [13]. In this context some researchers believe that clinical equipoise has been reached and are initiating randomised controlled trials to investigate whether SGLT2i could reduce cardiac surgery-associated acute kidney injury [14].

The primary objective of the study was therefore to assess the effect of preoperative SGLT2i exposure on cardiac surgery-associated acute kidney injury Stage $\geq$1 in an observational single-centre cohort of patients with a theoretical indication for long-term treatment. Secondary objectives were to assess the effect of preoperative SGLT2i exposure on cardiac surgery-associated acute kidney injury Stage 3, to assess the impact of SGLT2i discontinuation on cardiac surgery-associated acute kidney injury Stage $\geq$1 and to report the rates of adverse events associated with exposure to SGLT2i defined by urinary tract infections and euglycaemic keto acidosis.

Methods

Study Design

The study consisted of a single-centre retrospective cohort study of patients undergoing cardiac surgery in a cardiovascular and thoracic centre of a teaching hospital (France). The protocol was approved by the local ethics committee (Scientific and Ethical Committee of Hospices Civils de Lyon, France, institutional review board: 00013204). The latter waived the need for written informed consent, as the design was purely observational and the database anonymised. The data collection was approved by the national data protection commission (*Commission Nationale de l'Informatique et des Libertés*) and registered under identification 24-5025. The reporting of the study herein is in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines [15] (completed checklist in Supplementary Table 1).

Participants

Adults who had undergone cardiac surgery with cardiopulmonary bypass and who had $\geq$1 of the following indications for long-term treatment with SGLT2i [16] were included: diabetes associated with documented atheromatous disease or high cardiovascular risk (defined as age $\geq$55 years and the presence of $\geq$2 of the following factors: arterial hypertension, dyslipidaemia, obesity, albuminuria, current smoking); heart failure with a left ventricular ejection fraction <50%; or chronic kidney disease with a GFR <60 mL/min/1.73 m^2. Exclusion criteria were: surgery for cardiac transplantation or long-term mechanical circulatory assistance; preoperative cardiogenic shock; contraindication to long-term SGLT2i treatment (chronic kidney disease with an estimated GFR [eGFR] <20 mL/min/1.73 m^2, type 1 diabetes); non-indication for long-term SGLT2i treatment in chronic kidney disease (renal transplant, polycystic kidney disease, systemic lupus erythematous); acute kidney injury prior to surgery (as defined in the primary end point; if this criterion could not be met, patients were included only if the eGFR estimated by the Chronic Kidney Disease Epidemiology Collaboration formula was $\geq$75 mL/min/1.73 m^2); and death before postoperative day 1. The period of analysis was from January 2022 to February 2024; a short period to prevent significant changes in patient management over time, with the exception of heterogeneity in SGLT2i prescription [16–18] and SGLT2i perioperative management [10,11].

Data Handling

Data were retrospectively collected from computerised medical files using the Easily (V8.2.0.0, Lyon, France), Diane (V4.9.6.24758, Bow Medical, Lyon, France), and IntelliScape Critical Care and Anaesthesia (J.01.02 Philips, Andover, MA, USA) software.

Preoperative exposure to SGLT2i was defined as preoperative SGLT2i medication on prescriptions recorded in the medical file, and unexposure as no SGLT2i medication on prescriptions recorded in the medical file. Discontinuation of preoperative SGLT2i was defined as withdrawal $\geq$3 days before the surgery as recommended by the European Society of Cardiology guidelines [11]. No specific institutional protocol for perioperative SGLT2i management was in place during the study period, and management was left entirely to the discretion of the treating clinicians. Cardiac surgery-associated acute kidney injury Stage $\geq$1 was defined by a serum creatinine increase of 0.3 mg/dL within 48 hours or a serum creatinine $\geq$1.5 times baseline within 7 days or renal replacement therapy initiation within 7 days after the surgery, and cardiac surgery-associated acute kidney injury Stage 3 by a serum creatinine increase $\geq$3 times baseline or

≥4 mg/dL or renal replacement therapy initiation within 7 days after the surgery [13,19].

The preoperative data collected were: anthropometric data, medical history of arterial hypertension, chronic obstructive pulmonary disease, previous cardiac surgery, diabetes and insulin requirement, left ventricular ejection fraction (categorised as reduced if ≤40%, mildly reduced if 41%–49%, and preserved if ≥50%), coronaropathy (if any atheromatous was seen in coronarography), extra-cardiac arteriopathy (if any atheromatous was seen in vascular preoperative explorations), active endocarditis, aorta dissection, non-elective surgery, and preoperative critical state as defined by EuroSCORE II [20], medications (loop diuretics, renin angiotensin system inhibitors, and SGLT2i), haemoglobin, preoperative intra-aortic balloon pump, and creatinine values (the last preoperative value available in the medical file) with the calculation of preoperative eGFR (Chronic Kidney Disease Epidemiology Collaboration formula [21]). The simplified predictive index for renal replacement therapy was calculated as previously described [22]. Intraoperative data collected were surgery type (valvular, coronary artery bypass, combined and other), duration of cardiopulmonary bypass and aortic clamping, the use of an inotrope, labile blood product transfusions, the cardioplegia solution used, and the cumulative dose of norepinephrine administered in the operating room. The postoperative data collected were arterial lactate at day 1, creatinine values from day 1 to 7, the need for extra corporeal life support and renal replacement therapy, postoperative intensive care unit length of stay, in-hospital and intensive care unit mortality, euglycaemic ketoacidosis and urinary tract infection as reported by the treating clinician.

Statistical Analysis

Data distribution was estimated using skewness and kurtosis analysis, and the Shapiro–Wilk test. Data are expressed as mean±standard deviation, median [interquartile range] or count (%), as appropriate. Imputation for missing values was performed using a random forest trained algorithm after excluding postoperative creatinine values. Missing postoperative creatinine values were considered as missing not at random, mainly due to the transfer of patients without renal failure to surgical wards and were not imputed to not artificially create occurrence of the primary outcome. The sample size calculation when using inverse propensity score analysis requires the anticipated prevalence of treatment and the C-statistics of the propensity score model, which were unknown and difficult to estimate in the present study [23]. The treatment effect reported in recent cohorts is very large: risk ratio of 0.66 (95% confidence interval [CI] 0.57–0.75) in acute kidney injury occurrence in a diabetic general population; and 0.48 (95% CI 0.39–0.59) in contrast-induced nephropathy due to coronary procedure [6,7]. We therefore planned to include ≥100 patients in each group to provide unbiased estimations of treatment effect [24] and to estimate *a posteriori* the power of the study. For causal inference analysis, preoperative variables associated with the exposure (preoperative treatment with SGLT2i) were first identified by fitting a binomial generalised linear model with Lasso regularisation. The variables participating to the maximal reduction of the binomial deviance by five cross-validation were considered. Among these variables, those also associated with the outcome (cardiac surgery-associated acute kidney injury), based on a recent literature review [13], were then selected for logistic regression propensity score weighting. Two approaches were analysed. Inverse probability weighting that attributes a weight of 1/propensity score to exposed and 1/(1 − propensity score) to unexposed patients. If it allows the estimation of the average treatment effect in the whole population of the study, it is sensitive to extreme propensity scores and could fail to correct imbalance between groups. The overlap weighting method estimates the average treatment effect for the overlap population as exposed patients are weighted by the probability of not being exposed (1 − propensity score) and untreated patients are weighted by the probability of being exposed (propensity score). This method leads to exact balance between groups for the covariates used in the propensity score model, and improve precision of the average treatment effect without excluding study participants from the available sample but emphasises the comparison in a population with the most overlap in their characteristics [25,26]. Imbalance between groups was defined by a standardised mean difference between groups <0.1. The precision loss caused by propensity score weighting was expressed by the effective sample size [27] and the calculation of the variance inflation factor [23]. Estimation of the pairwise weighted average treatment effect of SGLT2i preoperative exposure on cardiac surgery-associated acute kidney injury was assessed by the calculation of the causal relative risk with its 95% confidence interval [95% CI], determined by the empirical sandwich variance formula [25]. To explore whether continuation of SGLT2i preoperatively impacted cardiac surgery-associated acute kidney injury, a subgroup analysis following the same approach was performed in patients with preoperative SGLT2i exposure. Adverse event rates potentially associated with preoperative SGLT2i were described without adjustment for comparison. Statistical analyses were performed using R version 4.3.3 (R Core Team, Vienna, Austria, 2017) and the packages *missForest* [28], *glmnet* [29], and *PSweight* [30]. All the tests were two-sided and a $p<0.05$ was deemed significant.

Results

Between 1 January 2022 and 29 February 2024, a total of 1,615 patients were screened for eligibility and 509 were included (Figure 1). A total of 117 patients were exposed to SGLT2i before surgery, among whom 65 (56%) did not discontinue treatment. Two hundred and seventeen patients

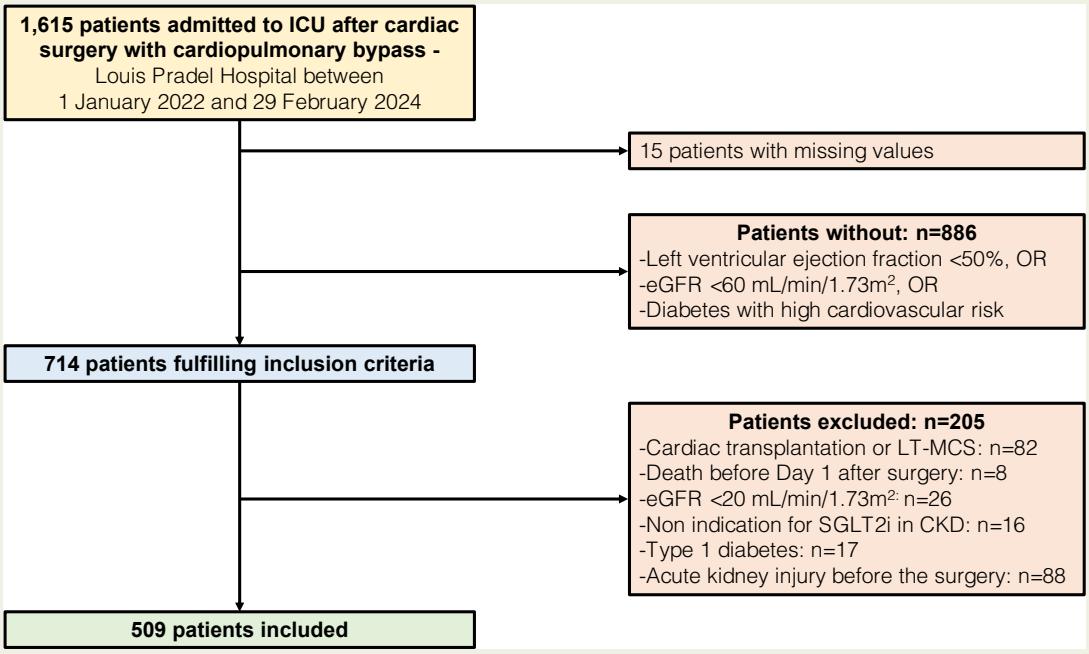


Figure 1 Study flow chart.
Abbreviations: CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; ICU, intensive care unit; LT-MCS, long-term mechanical circulatory support; SGLT2i, Sodium-glucose cotransporter-2 inhibitors.

(217) had diabetes associated with documented atheromatous disease or high cardiovascular risk, including 52 (24%) with preoperative SGLT2i exposure; 168 patients had left ventricular ejection fraction <50%, including 68 (40%) with preoperative SGLT2i exposure; and 175 patients had preoperative eGFR <60 mL/min/1.73 m^2, including 31 (18%) with preoperative SGLT2i exposure. Missing values are reported in Supplementary Figure 1. Variables associated with SGLT2i preoperative exposure and cardiac surgery-associated acute kidney injury included in the propensity score logistic regression model were: left ventricular ejection fraction, chronic arterial hypertension, chronic obstructive pulmonary disease, body mass index, preoperative haemoglobin, preoperative serum creatinine, loop diuretic, renin angiotensin system inhibitors, surgery, and simplified predictive index for renal replacement therapy (Supplementary Table 2). Common support of the propensity score was illustrated with a density plot (Supplementary Figure 2). The C-statistic for the propensity score model was 0.79 (95% CI 0.74–0.82). Inverse probability weighting did not lead to optimal balance between groups in baseline characteristics (Figure 2), hence only the overlap weighting propensity score method was used (effective sample size of 93 with preoperative exposure to SGLT2i and 214 patients without, variance inflation factor 1.21). The study had an *a posteriori* estimated power of 80% to detect a relative risk of 0.55 for cardiac surgery-associated acute kidney injury Stage ≥1 in the total population (and to detect a relative risk of 0.50 in the effective sample size). Baseline characteristics before and after overlap weighting are reported in Table 1. Intraoperative characteristics and postoperative outcomes after overlap weighting are reported in Table 2. The relative risk (preoperative exposure to SGLT2i versus no exposure) was 1.12 (95% CI 0.80–1.55, prevalence of 35% vs 30%) for cardiac surgery-associated acute kidney injury Stage ≥1, and 0.87 (95% CI 0.33–2.32, prevalence 4% vs 5%) for cardiac surgery-associated acute kidney injury Stage 3. In the subpopulation of patients with preoperative exposure to SGLT2i, the relative risk of cardiac surgery-associated acute kidney injury Stage ≥1 in patients who continued preoperative SGLT2i exposure, versus those who discontinued, was 0.95 (95% CI 0.57–1.58, prevalence 31% vs 31%; Figure 3, Supplementary Tables 3 and 4). Two surviving patients (2%) with preoperative exposure to SGLT2i presented euglycaemic ketoacidosis, one with and one without preoperative SGLT2i discontinuation. None of the patients presented euglycaemic ketoacidosis without preoperative exposure to SGLT2i. Urinary tract infections in the intensive care unit were reported in seven (6%) patients with preoperative exposure to SGLT2i (of whom four had no SGLT2i preoperative discontinuation) and 13 (3%) patients without preoperative exposure to SGLT2i (p=0.276).

Discussion

The main finding of the study is that we could not demonstrate that the preoperative exposure to SGLT2i is

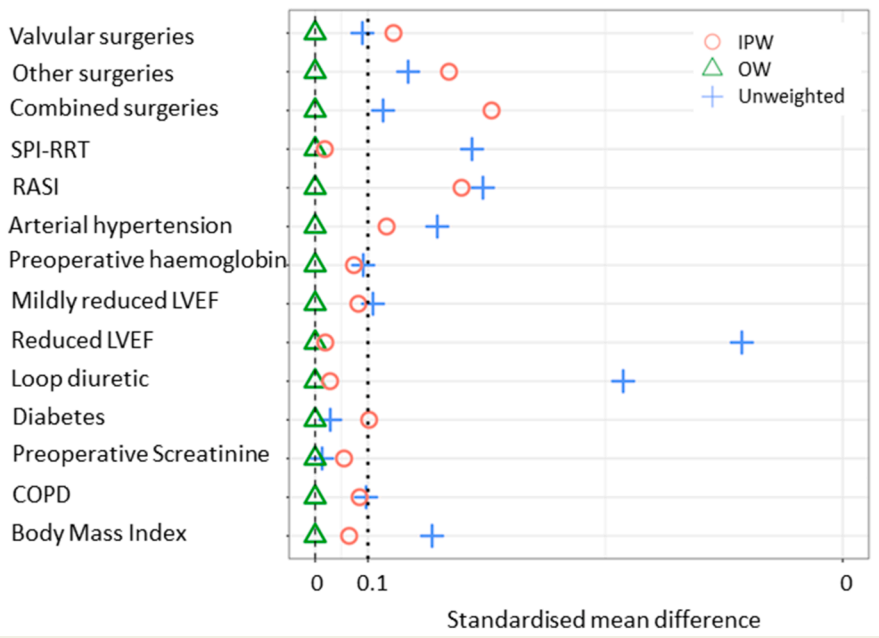


Figure 2 Balance between groups before and after inverse probability and overlap weighting.
Abbreviations: COPD, chronic obstructive pulmonary disease; IPW, inverse probability weighting; LVEF, left ventricular ejection fraction; OW, overlap weighting; RASI, preoperative renin angiotensin system inhibitors; Screatinine, serum creatinine; SPI-RRT, simplified prediction index for renal replacement therapy.

associated with a reduction in cardiac surgery-associated acute kidney injury in a population with a long-term treatment indication.

SGLT2i may prevent acute kidney injury through multiple pathways related to cardiorenal protection, inflammation, ischaemia-reperfusion injury, oxygen consumption, and vasoactive effects on glomerular haemodynamics. Their long-term renoprotective effects could also enhance kidney functional reserve prior to surgery [31]. Cardiac surgery-associated acute kidney injury is complex and multifactorial, resulting from hypoperfusion, ischaemia-reperfusion injury, inflammation, sympathetic activation, and direct nephrotoxicity, particularly in patients with reduced kidney functional reserve [13]. The pleiotropic effects of SGLT2i make them plausible candidates for renoprotection in this setting, and their potential benefits in medical or surgical contexts for acute kidney injury prevention have been suggested by observational studies [6,32,33]. However, SGLT2i could also contribute to acute kidney injury via mechanisms such as a decrease in GFR due to afferent arteriolar vasoconstriction mediated by tubuloglomerular feedback, arterial hypotension, or hypovolemia induced by osmotic diuresis [9]. These opposing effects on renal injury may help explain the differences observed between our results and those reported in contrast-induced acute kidney injury [7], non-specific [6] or non-cardiac postoperative acute kidney injury [33]. It is also possible that SGLT2i have no effect on certain key and specific determinants of cardiac surgery-associated acute kidney injury, such as those related to cardiopulmonary bypass, which potentially negating the renoprotective effects of SGLT2i in this specific setting. Another explanation may be that the study population is not limited to patients with type 2 diabetes, as most of the data on the nephroprotective effect of SGLT2i in the setting of acute kidney injury have been reported in this patient population [6,7]. Although several studies suggest that a similar effect could be observed regardless of diabetes status [34], and in critically ill patients [32].

The results could also be attributed to the design of the study. For instance, we could not clearly define the duration of preoperative exposure to SGLT2i from the medical files. In addition, there was heterogeneity in the population when considering preoperative SGLT2i discontinuation. In this context, the study results cannot be generalised to a population with a long-term preoperative exposure to SGLT2i and preoperative discontinuation (as recommended), nor to a population without preoperative treatment discontinuation. As renal metabolic impact seems to start in the first 3 days after initiation of SGLT2i, the effects on acute kidney injury could, however, be different in the two populations and may have impacted the results of the present study (even though the subgroup analysis does not strongly support this hypothesis [35]). In this context, it is difficult to anticipate how the results might have been different in a more homogeneous population considering preoperative duration and discontinuation of SGLT2. The exploratory subgroup analysis did not support an effect of treatment continuation. This result should not be overinterpreted, as

Table 1 Patient characteristics at baseline before and after overlap weighting.

Characteristic	Before overlap weighting			After overlap weighting		
	Not exposed to SGLT2i, n=392	Exposed to SGLT2i, n=117	SMD	Not exposed to SGLT2i	Exposed to SGLT2i	SMD
Anthropometric data						
Female, n (%)	112 (29)	29 (25)	0.086	28	28	0.016
Age, years, median [IQR]	69 [62–74]	66 [60–73]	0.163	69 [61–74]	68 [61–73]	0.018
Body mass index, median [IQR]	27 [24–30]	25 [23–28]	0.222	26 [23–29]	26 [23–29]	<0.001
Medical history						
Arterial hypertension, n (%)	236 (60)	57 (49)	0.232	53	53	<0.001
Chronic obstructive pulmonary disease, n (%)	18 (5)	8 (7)	0.097	6	6	<0.001
Diabetes, n (%)	182 (46)	56 (48)	0.029	49	49	<0.001
Coronaropathy, n (%)	274 (70)	94 (80)	0.243	72	80	0.179
Extra-cardiac arteriopathy, n (%)	201 (51)	58 (50)	0.034	53	51	0.033
Previous cardiac surgery, n (%)	48 (12)	17 (15)	0.067	15	15	0.009
Aortic dissection, n (%)	4 (1)	0 (0)	0.144	0	0	0.097
Active endocarditis, n (%)	23 (6)	6 (5)	0.032	6	5	0.042
Preoperative long-term medications						
Loop diuretic, n (%)	128 (33)	71 (61)	0.585	49	49	<0.001
Renin angiotensin system inhibitors, n (%)	241 (62)	89 (76)	0.319	70	70	<0.001
Insulin, n (%)	48 (12)	20 (17)	0.137	13	13	0.009
Preoperative left ventricular ejection fraction			0.816			<0.001
Reduced, n (%)	27 (7)	41 (35)		18	18	
Mildly reduced, n (%)	73 (19)	27 (23)		24	24	
Preserved, n (%)	292 (75)	49 (42)		58	58	
Preoperative laboratory data						
Haemoglobin, g/dL [IQR]	134 [121–145]	134 [122–145]	0.091	134 [121–145]	133 [121–144]	<0.001
Serum creatinine, mg/dL [IQR]	95 [78–117]	91 [78–110]	0.013	95 [78–117]	93 [79–110]	<0.001
eGFR, mL/min/1.73 m^2, median [IQR]	66 [55–92]	71 [56–90]	0.091	71 [55–92]	69 [55–90]	0.010
Surgery characteristics						
Type of surgery			0.300			<0.001
Valvular, n (%)	185 (47)	50 (43)		45	45	
Coronary artery bypass, n (%)	129 (33)	53 (45)		42	42	
Combined, n (%)	49 (13)	10 (9)		10	10	
Other, n (%)	29 (7)	4 (3)		3	3	
Simplified predictive index for renal replacement therapy, median [IQR]	2 [1–3]	2 [2–3]	0.297	2 [1–3]	2 [1–3]	<0.001
Preoperative critical state, yes, n (%)	8 (2)	4 (3)	0.085	3	4	0.071
Non-elective surgery, yes, n (%)	104 (27)	47 (40)	0.292	36	35	0.018

The characteristics of the targeted population are described with the weighted median [IQR] for the numerical and prevalence for the categorical baseline variables [26]. Effective sample size was 93 patients with preoperative exposure to SGLT2i and 214 patients without.

Abbreviations: SGLT2i, sodium-glucose cotransporter 2 inhibitor; SMD, standardised mean difference; IQR, interquartile range; eGFR, estimated glomerular filtration rate.

the analysis is probably severely underpowered due to the small sample size. In addition, the present study was only designed to detect a strong protective effect of pre-operative SGLT2i exposure on cardiac surgery-associated acute kidney injury. It therefore does not exclude a clinically significant effect, which is generally of a lower magnitude in this clinical context [36,37], as anticipated in the design of the future trials [14].

In the present study, we chose to report two adverse events observed with exposure to SGLT2i, euglycaemic ketoacidosis and urinary tract infections [10]. The former has been reported in several studies with a heterogeneous

Table 2 Intraoperative characteristics and postoperative outcomes before and after overlap weighting.

	Before overlap weighting			After overlap weighting		
	Non-exposed to SGLT2i, n=392	Exposed to SGLT2i, n=117	p-value	Non-exposed to SGLT2i	Exposed to SGLT2i	p-value
Intraoperative characteristics						
Cardiopulmonary bypass duration, min, median [IQR]	103 [81–132]	106 [85–132]	0.785	97 [81–123]	106 [85–131]	0.080
Aortic clamping time duration, min, median [IQR]	77 [60–99]	78 [63–101]	0.785	74 [58–93]	77 [63–104]	0.140
Inotrope, yes, n (%)	117 (30)	59 (50)	<0.001	37	47	0.209
Labile blood products transfusion, yes, n (%)	68 (17)	17 (15)	0.565	18	13	0.126
Norepinephrine, mg/kg median [IQR]	2.32 [1.40–4.14]	3.33 [1.60–5.02]	0.025	2.39 [1.37–4.22]	3.17 [1.49–5.02]	0.115
Postoperative outcomes						
Arterial lactate at day 1, mmol/L [IQR]	2 [1.5–2.6]	1.9 [1.5–2.8]	0.705	1.9 [1.5–2.8]	1.9 [1.4–2.8]	0.837
Postoperative mechanical circulatory support, yes, n (%)	13 (3)	12 (10)	0.005	4	8	0.159
Acute kidney injury grade 1 or more	137 (35)	36 (31)	0.468	30	35	NA*
Acute kidney injury grade 3	21 (5)	4 (3)	0.543	5	4	NA*
Renal replacement therapy	16 (4)	3 (3)	0.630	3	3	0.877
Intensive care length of stay, days [IQR]	2 [1–4]	3 [1–5]	0.131	2 [1–4]	3 [1–5]	0.236
Intensive care mortality, n (%)	11 (3)	9 (8)	0.034	2	7	0.017
In-hospital mortality, n (%)	26 (7)	13 (11)	0.161	7	12	0.155

p-values are from Mann–Whitney U tests for continuous and Pearson chi-square or Fisher's exact test for categorical variables. The characteristics of the targeted population are described with the weighted median [IQR] for the numerical and prevalence for the categorical variables [26].

*The comparative analysis after adjustment was performed using relative risk (see text). Effective sample size was 93 patients with preoperative exposure to SGLT2i and 214 patients without.

Abbreviations: SGLT2i, sodium-glucose cotransporter 2 inhibitor; IQR, interquartile range; NA, not applicable.

frequency in the perioperative period [38,39]. Although serious, it is a non-fatal complication if well-recognised by the clinician, which can occur even when treatment is stopped according to international guidelines [40]. However, further studies are needed to analyse its impact on clinically relevant outcomes. Considering urinary tract infections, we did not observe a significantly higher rate, but a previous retrospective cohort study suggested that SGLT2i use could be associated with a higher rate of perioperative urinary tract infection [41]. It is also of note that there was a significantly higher intensive care mortality in patients with preoperative exposure to SGLT2i. This result should be interpreted with caution, as the study and statistical analysis were not designed to assess this outcome, which was reported for information on effect and sample size estimation in future studies. This association was not statistically significant for in-hospital mortality. Exploratory outcomes did not differ in patients with preoperative exposure to SGLT2i, whether or not treatment was discontinued.

The present study has several limitations. As a single-centre study conducted in a period of heterogeneity in SGLT2i prescription and perioperative management in patients who underwent cardiac surgery with cardiopulmonary bypass, the generalisability of the results is limited. The use of a propensity score approach carries the risk of unmeasured confounders or insufficient adjustment for key population characteristics. If present, these factors may have biased the results against the SGLT2i-exposed population, as patients in this group had higher baseline severity indicators, including left ventricular dysfunction, lower GFR, and higher mortality. The small sample size and the heterogeneity in the definition of the exposure may have resulted in insufficient power to detect a clinically significant impact of preoperative exposure to SGLT2i on cardiac-associated acute kidney injury; however, it would have had the power to detect a strong difference mediated by SGLT2i exposure as observed in contrast-induced acute kidney injury. The results are also consistent with a larger study showing an association between preoperative exposure to SGLT2i and a reduced risk of postoperative acute kidney injury, except in the specific setting of cardiac surgery [33]. Another limitation regarding exposure is that we were unable to assess strict adherence to preoperative SGLT2i therapy or its discontinuation prior to surgery when required by the anaesthesiologist. Furthermore, the analysis focused on preoperative exposure and did not account for the postoperative reintroduction of SGLT2i, which could potentially have influenced creatinine variations during the 7-day postoperative period. To finish, the absence

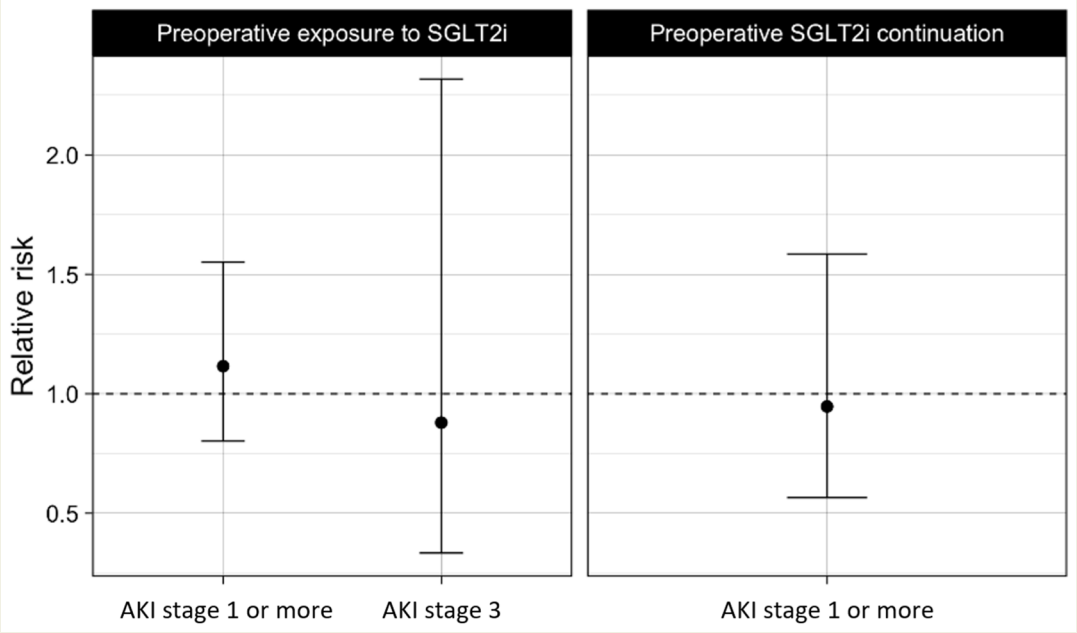


Figure 3 Causal relative risk of renal outcomes after cardiac surgery following preoperative SGLT2i exposure and continuation status. The relative risk (preoperative exposure to SGLT2i versus no exposure) was 1.12 (95% CI [0.80; 1.55], prevalence of 35 vs 30%) for cardiac surgery-associated acute kidney injury Stage ≥1, and 0.87 (95% CI [0.33; 2.32], prevalence 4 vs 5%) for cardiac surgery-associated acute kidney injury Stage 3. In patients with preoperative exposure to SGLT2i, the relative risk of cardiac surgery-associated acute kidney injury Stage ≥1 in patients who continued preoperative SGLT2i exposure, versus those who discontinued, was 0.95 (95% CI [0gs.57; 1.58], prevalence 31 versus 31%. Abbreviations: AKI, cardiac surgery-associated acute kidney injury; CI, confidence interval; SGLT2i, sodium-glucose cotransporter-2 inhibitors.

of diuresis-based acute kidney injury criteria may have biased the primary end point, although it remains controversial in the perioperative setting [42]. Several strengths can also be highlighted. The population was selected according to a theoretical indication for long-term treatment by SGLT2i to reduce the indication bias and respect the positivity assumption required in the propensity score analysis. In addition, a rigorous analysis plan for the causal inference analysis was used, from the selection of the variables used in the propensity score model to the choice of weighting approach to allow balance of the populations.

Conclusions

Preoperative exposure to SGLT2i was not associated with a significant reduction in cardiac surgery-associated acute kidney injury in the present retrospective cohort study. However, the study was underpowered to exclude a potential clinically significant effect of preoperative exposure to SGLT2i on cardiac surgery-associated acute kidney.

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Declaration of Competing Interests

There are no conflicts of interest to disclose.

Author Contributions

Study concept and design: A.V., C.B., J.L.F., M.J.L., and M.R. Acquisition of data: A.V, M.R., and Y.L.L. Interpretation of data: C.B., J.L.F., M.J.L., M.R., M.P., and N.M. Drafting of manuscript, statistical analysis: M.R. Study supervision: M.R. and M.J.L. Critical revision of the manuscript for important intellectual content: A.V., C.B., J.L.F., M.J.L., M.R., M.P., N.M., and Y.L.L.

Data Statement

All de-identified datasets may be available for secondary analysis upon reasonable request to the corresponding author.

Appendices

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.hlc.2026.01.004.

Use of Generative AI and AI-Assisted Technologies

Generative Pretrained Transformer version 4.0 (GPT-4.0) and GitHub Copilot have been used for syntax support in the R language. After using these tools, the author reviewed and edited the content as needed and take full responsibility for the content of the published article.

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